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1 lookDevTool_v001
2
3     "LookDevTool" is a Qt app for editing BldMtlGrp objects.
4
5     Its native file format is ".bmg" and is XML.
6
7     BldMtlGrp objects are used as input to the material
8     compiler, and represent a workspace that allows:
9
10        - selection and examination of input model(s)
11
12        - extraction of material assignment targets (MtlNode)
13
14        - creation of additional nodes in the assignment
15          heirarchy
16
17        - construction of material inheritance heirarchies
18
19        - alteration of the parent/child relationships in
20          the material or node heirarchies, using drag-n-drop
21
22        - control of which revision of the facility-wide
23          archetype material library is used as the root
24          of materials
25
26        - assignment of materials to material assignment targets
27          using drag-n-drop
28
29        - choice of shader version
30
31        - editing of material parameters
32
33     It has a standard QT app mainWindow with dockable widgets.
34
35     The central widget is not used yet, but would eventually
36     contain an OpenGL viewer for GTOs capable of showing
37     facesets and meshes (with an outliner in a splitter.)
38
39     It can also show a proxy sphere or patch with a material
40     render using simple fixed light rig to show a preview of
41     material parameters.
42
43     The material parameters are either
44        - locally set to a value
45        - inheritied from a parent
46
47     All parameters display themselves in one of the following
48     states:
49        - locally set
50        - set to a value by a parent
51        - unchanged from the default value from the SLO
52
53     Parameters can be grouped and their viewing properties
54     adjusted and saved with the workspace.
55
56     New materials defined locally can be checked into the
57     library with appropriate authentication.
58
59
60 Application Menus:
61
62     File
63
64         New                (create new BldMtlGrp object)
65         Open               (open saved .bmg file)

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66         Save                (save BldMtlGrp object as .bmg file)
67         -
68         Quit
69
70     Edit
71
72         Undo                (Undo in scope of app)
73         Redo
74         -
75         Copy
76         Cut
77         Paste
78         -
79         Show clipboard
80
81     View
82
83         BuildMtl            (create new BldMtlBrowser)
84         NodeDefn            (create new NodeDefnBrowser)
85         -
86         Compile              (create new compileWidget)
87         Checkin              (create new checkinWidget)
88         -
89         [ ] History          (Show/hide app history dockable widget)
90
91     Dialogs:
92
93         newBuildMtlDlg:
94             - choose library revision
95
96     Widgets:
97
98         - all are dockable, multiple instances possible
99
100    checkinWidget:
101        - Checkbox list of all mtl's
102        - Comments
103        - Buttons
104        - Success, fail pop-up showing loco ID
105    Menus:
106
107    compileWidget:
108        - Checkbox list of all mtl's
109        - Buttons
110        - log window
111    Menus:
112
113    Dockable Widgets:
114
115        Material Browser
116
117        Menus:
118
119            File
120                Import...
121                Export...
122
123            Edit
124                Undo
125                Redo
126                History
127                -
128                Copy
129                Cut
130                Paste

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131         -
132         Show clipboard
133
134     Select
135         None
136         All
137         Invert
138
139     View
140         [ ] Flat
141         -
142         [ ] Filter
143         Filters...
144         -
145         Local
146         Non-local
147         Library
148         External
149
150     Mtl
151         New
152         -
153
154     Node Browser
155
156
157 Application Actions:
158
159     newMtlGrp
160     openMtlGrp
161     saveMtlGrp
162     quit
163     newMtlBrowser
164     newNodeBrowser
165     newMtl
166     importMtl
167     exportMtl
168     compileSelected
169     compileAll
170     checkin
171
172
```